

CONVICTION ENGINE v5

Complete System Reference — Final Version

Flowchart · Plain English Guide · Scoring Architecture · Data Structures · Decision Tables · TELUS Example

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7. Data Structures	Python schema, Layer A–E, JSON fields	Ahil (coder)
8. Decision Tables	Full verdict mapping with all conditions	Rohit + Ahil
9. TELUS Worked Example	Complete scorecard with real numbers	Everyone
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Rohit Malhotra | To: Ahil Khan / Development Team

1. The Simplest Explanation — For Everyone

1.1 The Problem This Engine Solves

The lemonade stand analogy:

Imagine you own a street full of lemonade stands. Each morning a thermometer (your quant model) tells you which stands are getting hotter — selling more lemonade, busier, trending up. But a rising temperature doesn't tell you if the lemonade stand is actually a great business or just having a good week. The Conviction Engine is the doctor's report on the business itself. It answers: **is this a genuinely great business at a fair price?**

The quant model is ONE system containing many analytical functions. Each function is a rule that generates price-based BUY/SELL signals. The authoritative mapping below comes directly from FUNCTION_CONFIG in trade_arrival_analysis.py — the system's own orchestrator file.

Core functions registered in FUNCTION_CONFIG (trade_arrival_analysis.py):

Function name (FUNCTION_CONFIG key)	Source file	Fixed interval or user-selectable	Signal logic (from source code)
OSCILLATOR_DELTA	divergence_comp.py	INTERVAL_DAILY (fixed)	Stochastic/price divergence. Tracks price vs indicator divergence at daily resolution.
TRENDPULSE	trendline.py	All intervals (D/W/M/Q/Y user-selectable)	Trendline break signals. BUY/SELL when price breaks above/below computed trendlines.
BASELINEDIVERGENCE	general_divergence.py	INTERVAL_WEEKLY (fixed)	Price vs benchmark divergence on weekly candles. BUY: bullish divergence confirmed.
FRACTAL_TRACK	fib_ret.py	All intervals: D/W/M/Q (selectable)	Fibonacci retracement tracking. Signals at key Fibonacci levels (23.6%, 38.2%, 50%, 61.8%, 78.6%).
BAND_MATRIX	bollinger.py	All intervals (user-selectable)	Bollinger Bands. BUY: Close crosses above middle band from below. SELL: crosses below from above.
DELTADRIFT	distance.py	All intervals (user-selectable)	MACD distance. BUY: $MACD \leq$ lower threshold AND $EMA \text{ signal} \leq$ lower AND $RSI \leq 41$ AND MACD bullish.
SIGMASHELL	sigma.py	INTERVAL_DAILY (fixed)	Statistical sigma. BUY: $\sigma < -3$ AND price below 5-day ago price (mean-reversion signal).
PULSEGAUGE	sentiment.py	INTERVAL_WEEKLY (fixed for signals)	Sentiment/spread gauge. BUY: $\text{spread} > \text{BUY_THRESHOLD}$ AND $RSI < 50$. SELL: $\text{spread} > \text{SELL_THRESHOLD}$.
ALTITUDE_ALPHA	new_high.py	All intervals (user-selectable)	New highs breakout. BUY: price breaks above previous local maximum (altitude rising).
COMBINED_STRATEGY	combined_strategy.py	All intervals (user-selectable)	Composite: runs BAND_MATRIX + TRENDPULSE + DELTADRIFT together. Applies per-function success-rate filters.

Additional functions in codebase (not in main FUNCTION_CONFIG registry):

File	Fixed interval	What it does
fib_ext.py	INTERVAL_MONTHLY (fixed)	Fibonacci extension levels. Plots projected extension targets from major moves.
horizontal.py	All intervals (selectable)	Horizontal support/resistance levels. Identifies price levels where reversals cluster.
cycle_analysis.py	Various	Cycle trough detection. BUY at identified cycle troughs with higher-low and bullish phase conditions.
math_algo.py	D/W/M (selectable)	Statistical pair trading. Engle-Granger cointegration — Long/Short pairs when spread reverts.
fract_track_bands.py	INTERVAL_WEEKLY (fixed)	Fibonacci bands built from major upmoves ($\geq 20\%$). Used as input by combined_bands.py.

File	Fixed interval	What it does
combined_bands.py	INTERVAL_WEEKLY (fixed)	Merged FTrack bands + FStack bands + horizontal weekly levels into unified support/resistance map.

■ IMPORTANT — WHAT SHORT-TERM AND LONG-TERM SIGNALS MEAN IN THIS ENGINE:

When this document says 'short-term signal' or 'long-term signal', this refers to a signal fired by a quant model function running on a SHORT interval (e.g. interval='1d' — daily candles, produces signals that play out over days to weeks) versus a LONG interval (e.g. interval='1wk' or '1mo' — weekly/monthly candles, produces signals that play out over months). The signal_timeframe parameter in modify_signal() is set by the quant model function based on its own interval parameter. The conviction engine does NOT determine the timeframe — the quant model function determines it and passes it in.

1.2 How the Engine Works — Five Steps

Step 1 — Is this even an equity?

First check: is this a real company with financial statements? A currency pair (EUR/USD), an ETF (IGV), or a commodity future has no earnings, no moat, no CEO. The engine immediately returns NOT_APPLICABLE for those. It only works on individual company stocks.

Step 2 — What kind of business is it?

The engine auto-detects the business type from financial data — no human input needed. Four types: SaaS (software subscriptions), compounder (high-quality growing business), income (pipelines, utilities, telecoms — pays big dividends), cyclical (commodity/energy). Why does this matter? A pipeline at 4x debt/earnings is normal and healthy. A tech startup at 4x debt/earnings is dangerous. Same number, completely different meaning. The engine uses different measuring sticks for each type.

Step 3 — Score the business quality (BQ score)

15 questions about the business itself — not the price. Is it growing? Does it make real cash? Is the CEO good? Does it have a moat? Is the debt manageable? Each yes adds points, each no subtracts. This produces the BQ raw score: a number from about -14 to +22. KXS (Kinaxis) scores +14. TELUS scores -5.

Step 4 — Apply the valuation tax

Even a perfect business is a bad buy if you pay too much. The valuation tax is always negative or zero — it only reduces the score. A SaaS company trading at 10x its annual revenue needs two things to go right simultaneously: the business has to keep executing AND investors have to keep paying that high multiple. That double requirement earns a heavy tax. A pipeline at 4x revenue is normal — no tax. BQ + valuation tax = conviction_score.

Step 5 — Apply the FS quality check and yield trap

The FS check asks: is this business viable as a going concern? Even if the conviction score looks acceptable, if the business is genuinely deteriorating it caps how much you can buy. The yield trap asks: is that big dividend actually a warning sign? If a company's dividend yield has spiked far above its own history AND is above the normal level for its stock market — the engine cancels any buy, regardless of score. These are hard gates. The final number is the conviction_score.

1.3 What the Engine Outputs

conviction_score	What it means	BUY verdict	SELL verdict
≥ +8	Outstanding business at a good price	MAX CONVICTION — full size	PAUSE SELL — keep 70%
≥ +5	Good business, some caution needed	TACTICAL BUY — 75% size	PARTIAL EXIT — keep 50%
≥ +2	Acceptable, proceed carefully	REDUCED BUY — 40% size	FULL EXIT — clean out
< +2	Do not buy / exit quickly	CANCEL BUY	HARD EXIT
Yield trap fires	Dividend is a danger signal	CANCEL BUY (hard gate)	HARD EXIT (hard gate)

1.4 Short-Term vs Long-Term Signals — The Two Position Layers

The quant model can fire signals at different time horizons. The engine tracks two separate layers for every holding:

Layer	What it is	Concrete example (\$100k portfolio, 10% max in KXS = \$10k)	Opens when	Closes when
Core (long-term)	The structural thesis position — held for months to years	core_fraction = 0.75 → \$7,500 in KXS.TO. Placed when DeltaDrift fired a structural BUY.	Long-term signal fires BUY	Long-term signal fires SELL, OR conviction drops below +2
Tactical (short-term)	A trading layer added ON TOP of the core for shorter momentum moves	tactical_fraction = 0.25 → \$2,500 added when TrendPulse fired a momentum BUY. Total exposure: \$10,000. Tactical is removed when TrendPulse fires its exit — core stays.	Short-term signal fires BUY	Short-term signal fires SELL (core always untouched)

■ **Trailing-stop warning:** if a short-term BUY fires while the long-term position is approaching its stop level — the engine flags this and cancels the tactical add. You do not add exposure just as the structural trade is about to exit.

2. Master System Flowchart — End to End

Read top to bottom. Every signal goes through this exact sequence, in order.

STEP 0 — ASSET TYPE CHECK (first, before anything else)

Check yfinance info["quoteType"]

EQUITY → proceed to engine

ETF / INDEX / CURRENCY / CRYPTO / FUTURE → return NOT_APPLICABLE (no error thrown)

LAYER C — FULL QUARTERLY RECALCULATION (triggered, not scheduled)

Triggers: earnings_date_passed | manual_override | buyback_suspension | revenue_miss | mgmt_change | div_cut

1. Auto-detect business_type from yfinance sector/industry/payout data (manual override wins if set)

2. Pull yfinance: stock.info + quarterly_income_stmt + balance_sheet + cashflow (one full API set)

3. Compute 15 BQ dimensions — 7 auto from financials, 8 from manual overrides

4. Apply sector-calibrated scoring: Rule of 40 only for saas/compounder; distribution coverage for income;

WACC: income 5.5%, compounder 7.5%, cyclical 9%, saas 10%; net debt thresholds vary by type

5. Compute valuation tax with sector-specific EV/rev tiers and floor triggers

6. Build PE history using contemporaneous trailing-4Q EPS (NOT today's EPS vs history)

LAYER A — PERSISTENT JSON STORE (conviction_store/{TICKER}.json)

STATIC (quarterly): bq_raw, bq_components, roic_5y_avg, pe_20y_array, eps_ttm, fcf_ttm, net_debt_stored, fwd_revenue_stored, annual_div_per_share_stored, fd_votes, fd_direction, business_type, business_type_source (manual vs auto), high_beta_saas, manual_overrides

DYNAMIC (daily): price, pe_ttm, pe_percentile_20y, ev_fwd_rev, oey, div_yield,

every morning before market open

LAYER B — DAILY LIGHTWEIGHT RUNNER (ONE API call per ticker: fast_info price only)

Fetch: price + market_cap from fast_info (all other inputs read from stored JSON — no stock.info call)

Recompute: $pe_ttm = price / stored_eps$ | $pe_pctile = percentileofscore(stored_array, pe_ttm)$

$ev_fwd_rev = (mcap + stored_net_debt) / stored_fwd_rev$ | $oey = stored_fcf / mcap$

$div_yield = stored_annual_div / price$ | div_zscore vs stored 5Y mean/std

valuation_tax (price components) | $conviction_score = bq_raw + new_val_tax$

$fs_score = fs_quality_base + val_slice(oey, pe_pct, ev_rev)$ — OEY threshold by sector

$fs_class = classify(fs_score)$ | $yield_trap = market\text{-}calibrated\ two\text{-}condition\ check$

LAYER E — UNIVERSE RUNNER + DAILY REPORT GENERATOR

run_daily_universe(): calls daily_update() for all 200+ assets → alert_map {ticker:[flags]}

generate_daily_report(): universe health summary (strong/moderate/weak/conviction distribution)

on quant model BUY or SELL signal

HARD GATE: yield_trap = True?

BUY → CANCEL BUY | SELL → HARD EXIT (no score can override)

LAYER D — SIGNAL MODIFIER (reads store, <1 second, no API call)

1. Check signal_timeframe: 'long' (acts on core position) or 'short' (acts on tactical layer)

2. Apply FS graduated cap (Option B):

LONG signals: $fs=weak$ → conviction capped +1 (REDUCED BUY max)

$fs=moderate_low$ → conviction capped +4 (TACTICAL BUY max)

SHORT signals: $fs=weak$ → conviction capped +2 | $fs=moderate_low$ → no cap

3. Trailing-stop check: if short BUY + long position near stop → CANCEL tactical add

OUTPUT — SignalModification (verdict + sizing + position layers + rationale)

BUY: MAX CONVICTION (100%) | TACTICAL BUY (75%) | REDUCED BUY (40%) | CANCEL BUY (0%)

SELL: PAUSE SELL (retain 70% core) | PARTIAL EXIT (50%) | FULL EXIT | HARD EXIT

fd_direction adjusts sizing within tier | position_layers updated and persisted

3. Key Terminology — Every Term Defined

ARR — Annual Recurring Revenue

For SaaS companies only: the annualised value of all active subscription contracts. Excludes one-time fees and services. When we say 'ARR grew 14%→18%→20%', we mean the subscription base grew at those rates sequentially — it is accelerating. NOT to be confused with Annual Run Rate (a looser general business term).

ARR acceleration

The RATE of ARR growth is increasing — not just positive. 14%→18%→20% is accelerating (each period faster). 20%→15%→12% is decelerating (even though all positive and high). This is the second derivative of revenue. The engine checks this via the revenue_accelerating manual flag and the fd_direction vote.

Trailing quarter ARR rate

The ARR growth rate at the END of a quarter — the velocity carrying into the NEXT period. KXS exited Q4 2025 at 20% ARR growth. That is the momentum entering Q1 2026. NOTE: 'exit' here means end-of-period. It does NOT mean a trade exit signal.

conviction_score

The single final number. BQ_RAW + valuation_tax (then FS cap applied). Range in practice: -12 to +10. This is the quant model's fundamental modifier — one number, one decision.

BQ_RAW (Business Quality Raw Score)

Sum of 15 quality dimensions. No valuation. Answers: how good is this business intrinsically? Range: -14 to +22. KXS scores +14. TELUS scores -5.

valuation_tax

Always ≤ 0. Applied to BQ to produce conviction_score. Sector-calibrated: SaaS floor at 4x EV/rev, income floor at 6x, compounder at 3x, cyclical has no floor.

OEY — Owner Earnings Yield

FCF / Market Cap. If you bought 100% of this business today, what annual cash yield would you earn? Buffett's primary valuation check. Sector-calibrated thresholds: saas ≥5% = strong, compounder/cyclical ≥4%, income ≥3%. Below 1% = you are pricing in far-future earnings with no margin of safety. Compare: 10Y US Treasury ~4.2% risk-free. A business at 5% OEY pays more AND can grow that yield over time.

fs_score (Fundamental Strength)

0–100 composite = fs_quality_base + daily valuation slice (OEY, PE pctile, EV/rev). Buckets into: strong / moderate_high / moderate / moderate_low / weak. Used to cap conviction_score before verdict mapping (Option B graduated cap).

fs_quality_base

Previously called fs_score_ex_valuation. Simply BQ_RAW rescaled: 50 + (BQ × 2.5). BQ +10 → base 75. BQ -6 → base 35. Computed quarterly, static between earnings.

PE percentile (20Y history)

Where today's PE sits in 20 years of this stock's own PE history. Self-calibrating — no sector adjustment needed. 85th percentile = near bubble peak. 15th = crisis-level cheap. Built using CONTEMPORANEOUS trailing-4Q EPS. Using today's EPS vs history is wrong.

fd_direction (Fundamental Direction)

5 direction votes: revenue / margins / FCF / EPS revision / buyback direction. Each positive/stable/negative. Majority wins. Positive fd_direction → bumps sizing up within tier (TACTICAL BUY goes from 75% to 85%). Negative → pulls sizing down (75% to 60%).

divergence_signal

Price ≥15% below 52-week high for ≥60 days AND fd_direction ≠ negative. Auto-detected. The 52W high is the most recent market consensus high-water mark. Price below it while fundamentals are stable = market is mispricing the business. The 60-day filter prevents fresh breakdowns from triggering falsely. +2 to BQ score.

business_type

Auto-detected from yfinance sector/industry/payout data. Manual override wins if set. Four types: saas | compounder | income | cyclical. Drives: WACC rate, OEY threshold, net debt/EBITDA thresholds, EV/rev tax tiers, Rule of 40 applicability, margin_quality substitution.

WACC (sector-calibrated)

Weighted Average Cost of Capital — the rate at which a business must earn on new investment to create value.

Sector-specific: income 5.5% (regulated, low risk), compounder 7.5%, cyclical 9%, saas 10% (rate-sensitive, far-future cash flows). Universal 8% WACC misprice utilities (7% ROIC looks like destruction vs 8% WACC; correct: +1.5% spread vs real 5.5% WACC).

distribution_coverage_ratio

For income businesses (telecoms, pipelines, utilities): FCF / annual dividend obligation. REPLACES Rule of 40 for income type in the margin_quality BQ dimension. >2.0x = healthy. ≥1.2x = sustainable. <1.0x = dividend is partly debt-funded = danger. TELUS at 0.87x = income type correctly penalised here.

yield_trap

Fires when BOTH: (1) dividend yield is >1.5 std deviations above this stock's own 5Y mean, AND (2) yield exceeds market-absolute threshold (NZ: 12%, AU: 10%, CA: 7%, UK: 9%, US/India: 6%). Two conditions required — prevents false positives on high-yield markets like NZX. Hard gate: CANCEL BUY or HARD EXIT, no override possible.

core_fraction / tactical_fraction

Position layer fractions. core_fraction = long-term structural position (months to years). tactical_fraction = short-term trading layer stacked on top. Example: max position \$10,000. core=0.75 → \$7,500 structural. tactical=0.25 → \$2,500 trading. A short-term SELL zeroes tactical_fraction only. Core stays unless long-term signal exits.

signal_timeframe

Parameter passed to modify_signal(): 'long' or 'short'. This reflects the INTERVAL of the quant model function that fired the signal — NOT a category of function. A Bollinger Bands function running on daily candles (interval='1d') produces a short-term signal (signal_timeframe='short') acting on the tactical layer. The same Bollinger Bands function running on monthly candles (interval='1mo') produces a long-term signal (signal_timeframe='long') acting on the core layer. The quant model function sets this — the conviction engine receives it as input.

long_position_near_stop

Boolean passed to modify_signal() by the quant model. Set True when the long-term position is approaching its trailing stop or trend-exit level. Engine flags and CANCELS any short-term tactical BUY add — prevents adding exposure at the worst possible moment (just before the structural trade exits).

10Y_Resilience

Binary label from the original V3 brief. 'TOP 10% (Moat Intact)' if ROE > 15%. Stored in record for reference. Quick proxy check.

Margin of Safety

= $(1 - \text{PE_TTM} / 25) \times 100\%$. Percentage discount vs 25x PE premium baseline. Positive = trading below premium. Negative = trading above. Stored in record.

4. Scoring Architecture

4.1 Three-Stage Formula

Stage	Name	Range	Frequency	Core question
1	BQ_RAW	-14 to +22	Quarterly	How good is this business intrinsically?
2	valuation_tax	-10 to 0	Daily (price)	How much does entry price compress returns?
3	conviction_score = BQ + tax (then FS cap applied)	-12 to +10 (practical)	Daily	Do fundamentals support this trade at this price?

4.2 BQ Score — 15 Dimensions

Seven computed automatically from financial data. Eight require manual qualitative overrides. The margin_quality dimension is type-substituted: Rule of 40 for saas/compounder, distribution coverage for income, normalised margin for cyclical.

#	Dimension	Source	Range	Investor basis	What it measures
1	revenue_quality	Auto	-2→+2	General	Gross margin, FCF conversion ≥90%, net retention ≥95%, RPO growth
2	growth_trajectory	Auto+manual	-2→+2	General	Revenue growth level + is the RATE accelerating (2nd derivative)?
3	margin_quality (TYPE-SUBSTITUTED)	Auto	-2→+2	Ackman+Burry	SaaS/compounder: Rule of 40 (rev_growth% + FCF_margin%). Income: distribution coverage (FCF/annual_div — how covered is the dividend?). Cyclical: normalised FCF margin vs cycle.
4	balance_sheet	Auto	-2→+2	General	Altman Z + net debt/EBITDA with SECTOR-CALIBRATED thresholds (see Section 5)
5	roic_wacc_spread	Auto (5Y avg)	-1→+2	Buffett/TCI	5Y avg ROIC minus sector-calibrated WACC. Positive = value creation per \$ reinvested
6	gross_margin_trend	Auto	-1→+1	TCI	Gross margin change over 3 years. Expanding ≥200bp = genuine pricing power
7	debt_maturity_risk	Auto	-1→0	Burrry/Silver Pt	Short-term debt >50% of total AND net lev above cyclical safe threshold
8	ceo_quality	Manual 0-10	-1→+2	General	Leadership quality + domain expertise. New CEO = additional -1 transition penalty
9	mgmt_capital_allocation	Manual 0-10	-1→+2	Buffett	SEPARATE from CEO. M&A; track record, buyback timing, dividend discipline
10	competitive_moat	Manual 0-10	-1→+2	Buffett/Munger	Gartner rank, proprietary IP, switching costs, customer lock-in
11	macro_tailwind	Manual -1/0/1/2	-1→+2	General	Net macro environment. KXS: tariff chaos = direct demand driver = +2
12	divergence_signal	Auto-detect+confirm	0 or +2	Price vs fundamentals	Price ≥15% below 52W high for ≥60 days AND fd_direction ≠ negative
13	deal_delay_risk	Manual bool	-1→0	General	Large enterprise deal timing slippage flagged in latest quarter
14	insider_ownership	Manual float	-1→+2	Buffett/Burrry	>15% = founder-level alignment. <1% = management treating it as a job
15	reinvestment_runway	Manual float	-1→+1	Ackman/TCI	TAM/current revenue. <3x = running out of room to compound at high ROIC

BQ_RAW = sum of all 15. Max achievable ≈ +22 across ALL business types (type-substitution in dimension 3 ensures no type is disadvantaged on scale).

4.3 FS Score and Graduated Cap (Option B)

fs_quality_base = 50 + (BQ_RAW × 2.5). Static between earnings. **fs_score** = fs_quality_base + daily valuation slice (OEY, PE pctile, EV/rev).

fs_score	fs_class	Cap on conviction — LONG signals	Cap on conviction — SHORT signals
≥75	strong	No cap	No cap
≥55	moderate_high	No cap	No cap
≥40	moderate	No cap	No cap
≥25	moderate_low	conviction = min(score, +4) — TACTICAL BUY max	No cap
<25	weak	conviction = min(score, +1) — REDUCED BUY max	conviction = min(score, +2)

FS is applied as a POST-COMPUTATION cap on conviction_score. Flow: compute conviction → apply FS cap → apply yield trap hard gate → map to verdict. It is NOT a pre-filter. Short signals get relaxed caps because a tactical trade does not require the same quality bar as a structural hold.

5. Sector Calibration — Why Numbers Differ by Industry

The same financial ratio means completely different things in different industries. The engine uses different measuring sticks for each business type. Using universal thresholds creates systematic errors for non-SaaS businesses.

5.1 OEY Thresholds

business_type	OEY strong threshold	OEY expensive threshold	Rationale
saas	≥5%	<1%	Asset-light. Minimal capex. FCF ≈ true owner earnings.
compounder	≥4%	<1%	Moderate capex. Still predominantly cash-generative.
cyclical	≥4% (mid-cycle only)	<0.5%	Single-year OEY misleads at peak cycle.
income	≥3%	<1%	Heavy regulated capex. Lower FCF yield is structurally acceptable.

5.2 Net Debt/EBITDA Thresholds

business_type	Safe threshold	Concern threshold	Danger threshold	Rationale
saas/compounder	Net cash (≤0x)	≤1.5x	≥5x	Standard — asset-light, uncontracted cash flows
income (pipelines/utilities)	Net cash	≤2.5x	≥7x	Contracted cash flows service higher leverage. Pembina/TELUS at 3x is manageable.
cyclical	Net cash	≤1.0x	≥3.5x	Commodity EBITDA can halve in a downturn. Same leverage is far more dangerous.

5.3 WACC (Weighted Average Cost of Capital)

The ROIC/WACC spread scores whether a business creates or destroys value with every dollar it reinvests. Using a universal 8% WACC misprice regulated utilities — a utility at 7% ROIC looks like a value destroyer vs 8% WACC, but against its real 5.5% WACC it creates +1.5% spread — healthy for a regulated asset.

business_type	Assumed WACC	Rationale
income	5.5%	Regulated returns, contracted cash flows, low risk premium
compounder	7.5%	Moderate risk, stable but uncontracted revenues
cyclical	9.0%	Commodity/cycle risk, earnings volatility
saas	10.0%	Rate-sensitive, far-future cash flows, high risk premium

5.4 EV/Revenue Valuation Tax Tiers

A pipeline at 4x EV/rev is structurally normal. A SaaS company at 4x EV/rev is moderately valued. A cyclical miner at 4x EV/rev is expensive. Different floor triggers per type.

business_type	Tier 1 (-1)	Tier 2 (-2)	Tier 3 (-3)	Tier 4 (-4)	Floor trigger (min -5)
saas	≥3x	≥5x	≥8x	≥12x	≥4x
compounder	≥1.5x	≥3x	≥6x	≥8x	≥3x
income	≥4x	≥6x	≥8x	≥10x	≥6x
cyclical	≥1x	≥2.5x	≥4x	≥5x	None (no floor)

5.5 Rule of 40 and Margin Quality Substitution

business_type	margin_quality dimension uses	Rationale
saas	Rule of 40 (rev_growth% + FCF_margin%)	Designed for SaaS. ≥ 40 = top tier. Measures growth + profitability simultaneously.
compounder	Rule of 40	Applicable to growing cash-generative businesses.
income	Distribution coverage ratio (FCF / annual_div_obligation)	The single most important income stock metric. $< 1.0x$ = dividend is debt-funded. REPLACES Rule of 40 which would give a pipeline a meaningless or false score.
cyclical	Normalised FCF margin vs cycle proxy	Rule of 40 not applicable. Checks if FCF margin is positive and revenue growing (healthy part of cycle) vs negative/contracting (trough).

6. Position Layers and Timeframe Architecture

6.1 The Core Problem This Solves

The quant model is ONE system. Its authoritative function list and interval mapping comes from `FUNCTION_CONFIG` in `trade_arrival_analysis.py` — the orchestrator. 10 functions are registered in the main pipeline. 6 additional functions exist in the codebase (see Section 1 for full table).

Key insight for Ahil: **`trade_arrival_analysis.py` is already the signal dispatch layer**. It runs all registered functions, collects their signals into `signallist[]`, and applies success-rate filters. The conviction engine call should plug INTO this layer — not into each individual function file. One integration point, not 10.

The fix: every signal now carries a **`signal_timeframe`** parameter ('long' or 'short'). Long signals act on the structural core position. Short signals act on a separate tactical layer. They are tracked independently and close independently.

6.2 Four Conflict Scenarios and How the Engine Handles Each

Scenario	Situation	Engine behaviour
Short SELL vs open long	TrendPulse fires 3-week overbought exit. DeltaDrift long still active.	Close tactical layer (take profit on \$2,500). Core \$7,500 untouched. Verdict: FULL EXIT or PARTIAL EXIT on tactical layer only.
Short BUY, long still open	Momentum bounce. Long-term thesis unchanged. Conviction still $\geq +5$.	Allow tactical add with flag 'TACTICAL ADD to existing core position'. Sum (core + tactical) capped at 1.0 of max position size.
Short BUY, conviction deteriorated	TrendPulse fires BUY. But since the core was opened, conviction has dropped to $+1$.	Engine flags 'TACTICAL_ADD_WARNING: conviction deteriorated'. Tactical sizing automatically halved. Analyst should review core thesis.
Short BUY, long near trailing stop	Bounce signal fires. But quant model has flagged long-term position near its stop.	Engine fires TRAILING_STOP_WARNING and CANCELS the tactical add entirely. Do not add exposure just as the structural trade is about to exit.
Long SELL fires	Structural thesis has ended. Core position exits per conviction verdict.	Core exits per PAUSE/PARTIAL/FULL/HARD schedule. Tactical layer also closed — if structural case weakens, tactical must close too.

6.3 Position Layer Concrete Example — \$100,000 Portfolio

Maximum intended position in KXS.TO: 10% of portfolio = **\$10,000**.

Event	Action	core_fraction	tactical_fraction	\$ in KXS
DeltaDrift fires long-term BUY. Conviction +6, <code>fd_direction=positive</code> → TACTICAL BUY 85%	Set core = 0.85	0.85	0.00	\$8,500
TrendPulse fires short-term BUY. Conviction +6 → allow tactical add	Set tactical = 0.15 (cap: 1.0 - 0.85)	0.85	0.15	\$10,000 (fully sized)
TrendPulse fires short-term exit. Conviction still +6.	Zero tactical only. Core untouched.	0.85	0.00	\$8,500
DeltaDrift fires long-term SELL. Conviction +6 → PARTIAL EXIT retain 50%	Core $\times 0.50$. Tactical already 0.	0.43	0.00	\$4,300
Conviction drops to -3 (deteriorating fundamentals). Long SELL fires.	HARD EXIT. Core to zero. Tactical to zero.	0.00	0.00	\$0

6.4 Integration Pattern — Any Quant Model Function

Every quant model function passes its own interval to determine `signal_timeframe`. The conviction engine does not infer this — it receives it as a parameter.

```
# Pattern: any quant function fires a signal → call modify_signal with its interval context
# Bollinger Bands on daily candles (interval='1d') → short-term, tactical layer:
```

```
result = modify_signal('KXS.TO', 'BUY',
signal_timeframe='short',
quant_model_name='BollingerBands_1d_period20',
signal_strength=0.75)

# Same function on monthly candles (interval='1mo') → long-term, core layer:
result = modify_signal('KXS.TO', 'BUY',
signal_timeframe='long',
quant_model_name='BollingerBands_1mo_period20',
signal_strength=0.80)

# Check if long position is near trailing stop before adding tactical layer:
result = modify_signal('KXS.TO', 'BUY', signal_timeframe='short',
quant_model_name='BollingerBands_1d',
long_position_near_stop=True) # engine will CANCEL tactical add
```

7. Data Structures — Technical Architecture

7.1 JSON Record Schema — conviction_store{TICKER}.json

Zone	Key fields	Written by	Read by	Freq
STATIC FUNDAMENTALS	bq_raw, bq_components, roic_5y_avg, fs_quality_base, pe_20y_array, eps_ttm, eps_fwd, fcf_ttm, net_debt_stored, fwd_revenue_stored, annual_div_per_share_stored, fd_votes, fd_direction, manual_overrides, business_type, business_type_source (manual/auto), high_beta_saas, buyback_history, eps_revision_history, 10Y_Resilience, Margin_of_Safety	full_recalculation()	daily_update() modify_signal()	Quarterly
DYNAMIC SLICE	price, market_cap, enterprise_value, pe_ttm, pe_percentile_20y, ev_fwd_rev, owner_earnings_yield, dividend_yield_current, dividend_yield_zscore, conviction_score, fs_score, fs_class, yield_trap_warning, flags[], last_flag_date	daily_update()	modify_signal() report()	Daily
POSITION LAYERS	position_layers{core_fraction, tactical_fraction, core_signal_date, tactical_signal_date, core_model, tactical_model}	modify_signal()	modify_signal() report()	On signal
METADATA	ticker, last_full_calc, full_recalc_trigger, next_full_calc_trigger, last_daily_update, last_override_update, auto_divergence_detected, buyback_suspension_flag, yield_trap_mkt_threshold	Both	Reporting	As updated

7.2 Layer Function Map

Layer	Function(s)	API calls	Inputs	Outputs	Side effects
A (Store)	load_record() save_record()	0	ticker	dict None	R/W JSON
B (Daily)	daily_update(ticker, record, force)	1 (fast_info)	ticker, record	(record, flags[])	Saves JSON
C (Quarterly)	full_recalculation(ticker, overrides, trigger)	Full yfinance	ticker, dict, str	complete record	Saves JSON; Macrotrends if thin PE
D (Signal)	modify_signal(ticker, signal, timeframe, ...)	0 (store read)	ticker, signal, timeframe, strength	SignalModification	Updates position_layers; saves JSON
E (Universe)	run_daily_universe(tickers)	1 per ticker	list[str]	alert_map dict	Calls B for each
Report	generate_daily_report(alert_map)	0	alert_map	formatted str	Prints to stdout
Calendar	check_earnings_triggers()	0 to check; full recalc if due	overrides map opt	list triggered	Fires C on passed dates
Init	initialise_universe(tickers, overrides)	Full per ticker	list, map	results dict	Day 0: calls C for each
Overrides	update_overrides(ticker, updates, recompute)	0	ticker, dict	updated record	Updates JSON; optional BQ recompute

7.3 Key Design Decisions

Why flat JSON (not SQLite)? 200 assets × 3KB = 600KB. Trivially small. Human-readable, version-controllable, zero infrastructure. At 2000 assets (~6MB) swap load_record/save_record for cursor.execute() — same schema.

Why separate static from dynamic? stock.info takes 500ms–2s per ticker. 200 tickers × stock.info = 100–400 seconds pre-market. Unacceptable. By caching fundamentals quarterly, daily runner needs only fast_info for price: 100ms × 200 tickers = 20 seconds total.

Why ≥ 20 PE data points before Macrotrends supplement? With 8 points, percentile estimate error = ± 25 percentage points. At 20 points (~5Y quarterly), error drops to ± 12 pp — acceptable. Macrotrends: US-only, ~52 points (13Y), free, no API key. Non-US: Gurufocus free tier or manual JSON entry.

8. Complete Decision Tables

8.1 BUY-Side Verdict Mapping

conviction_score (after FS cap)	Verdict	Base sizing	fd=positive	fd=negative	Notes
≥+8	MAX CONVICTION	100%	100%	100%	yield_trap → CANCEL BUY (hard gate)
≥+5	TACTICAL BUY	75%	85%	60%	yield_trap → CANCEL BUY
≥+2	REDUCED BUY	40%	50%	25%	yield_trap → CANCEL BUY
<+2	CANCEL BUY	0%	0%	0%	yield_trap also → CANCEL BUY
Any	CANCEL BUY	0%	—	—	FS=weak (long): always CANCEL BUY regardless

8.2 SELL-Side Verdict Mapping

For long signals: sizing_pct = fraction of CORE position to retain (0 = full exit). For short signals: closes tactical layer only regardless of verdict name.

conviction_score	Verdict (long signal)	Retain (core)	Short signal behaviour
≥+8	PAUSE SELL	70%	Close tactical only. Core untouched.
≥+5	PARTIAL EXIT	50%	Close tactical only. Core untouched.
≥+2	FULL EXIT	0%	Close tactical. Core to 0%.
<+2	HARD EXIT	0%	Close both. Exit quickly.
Yield trap	HARD EXIT	0%	Hard gate — both layers closed.

8.3 FS Cap by Timeframe

fs_class	Long signal — cap	Short signal — cap	Rationale
strong	No cap	No cap	Full conviction applies
moderate_high	No cap	No cap	Full conviction applies
moderate	No cap	No cap	Normal — proceed
moderate_low	min(conviction, +4)	No cap	Structural hold needs quality bar; tactical trade less so
weak	min(conviction, +1)	min(conviction, +2)	Deteriorating business. Short: small tactical ok. Long: almost never buy.

8.4 Asset Type Gate — First Check

Asset	yfinance quoteType	Engine output
Individual equity (stock)	EQUITY	Full BQ / FS / conviction calculation
Sector ETF (IGV, SPY etc)	ETF	NOT_APPLICABLE — ETF financials are fund-level aggregates, not operational
Index	INDEX	NOT_APPLICABLE
FX pair (EUR/USD)	CURRENCY	NOT_APPLICABLE — no fundamental modifier for currencies
Cryptocurrency	CRYPTOCURRENCY	NOT_APPLICABLE
Commodity future	FUTURE	NOT_APPLICABLE
Mutual fund	MUTUALFUND	NOT_APPLICABLE

9. TELUS Worked Example (T.TO)

TELUS is the reference name for the income business type. It demonstrates the sector calibration working correctly — different thresholds, different dimension substitution, yield trap firing.

9.1 Key Financials (2024 Annual Report + Apr 2026)

Metric	Value	Context
Revenue	CAD \$20.4B	Mature Canadian telecom
EBITDA	CAD \$8.2B	Solid but capex-intensive
Net Income	CAD \$0.74B	Suppressed by D&A; on \$50B asset base
FCF	CAD \$1.6B	After heavy 5G/fibre capex spend
Total Debt	CAD \$27.5B	Funded 5G/fibre rollout
Net Debt / EBITDA	3.28x	Income safe $\leq 2.5x$ — marginally elevated
Dividend Yield	7.3% (~\$1.61/share)	TSX market absolute threshold: 7%
FCF Dividend Coverage	0.87x	Dividend is 87% covered — gap funded by debt or asset sales
Payout Ratio	220% of net income	Remainder covered by D&A; cash + borrowing
ROIC (5Y avg)	5.5%	Exactly at income WACC 5.5% — zero value creation spread
EV / Fwd Revenue	~2.5x	Income tier 1 = 4x → no entry multiple penalty
Altman Z	~1.4	Below 1.8 = distress zone
Auto-detected type	income	Detected via: yield 7.3%>2.5%, payout 220%>45%, industry='telecom'

9.2 BQ Score Breakdown

Dimension	Score	Notes (income sector — calibrated thresholds)
revenue_quality	+1	Gross 39% ok for telecom; FCF conv 2.16x (D&A-driven; high); retention 88% < 95%
growth_trajectory	0	Revenue +3.2% — mature saturated Canadian telecom market
margin_quality (INCOME TYPE: distribution coverage)	-1	FCF coverage 0.87x < 1.0x = dividend partially debt-funded. Rule of 40 NOT applied (income type substitution)
balance_sheet	-1	Net debt/EBITDA 3.28x (income safe $\leq 2.5x$ = neutral). Altman Z 1.4 < 1.8 = -1 penalty
roic_wacc_spread	0	ROIC 5.5% vs income WACC 5.5% = 0% spread. Exactly at cost of capital — not creating value
gross_margin_trend	-1	Telecom margins structurally compressing — more competition, fibre opex
debt_maturity_risk	0	Long-dated bonds — near-term refinancing risk manageable despite high leverage
ceo_quality	-1	Score 4/10 + new CEO transition penalty. Entwistle retired after 24 years.
mgmt_capital_allocation	-1	Score 3/10 — aggressive leverage build; 5G ROI timeline uncertain
competitive_moat	0	Score 4/10 — regulated duopoly but weak vs Rogers on customer loyalty
macro_tailwind	-1	Rate sensitivity: \$27B debt. Every 100bp rate move = ~\$270M incremental interest cost
divergence_signal	0	Price fell AND fundamentals weakened simultaneously. Not a divergence setup.
deal_delay_risk	0	N/A for telecom subscription
insider_ownership	0	~1% management ownership — treating it as a job, not an investment
reinvestment_runway	-1 → 0	3x TAM/rev (mature market) → scores 0 per calibration
BQ_RAW TOTAL	-5	

9.3 Valuation Tax and Conviction

Component	Score	Notes
entry_multiple (income tiers: 4/6/8/10x)	0	EV/rev 2.5x below tier 1 (4x) — no penalty for income
pe_hist_percentile	-2	PE 34x at ~82nd pctile of own history — near high end
growth_multiple_fragility	0	Revenue growth 3.2% — not fragile at this multiple
business_type relief (+1 for cheap income)	+1	EV/rev below tier 1 → income gets modest relief
deal_delay_signal / market_regime_beta	0	N/A
VALUATION TAX TOTAL	-1	No -5 floor — income floor trigger is 6x; EV/rev is 2.5x

	Value	
BQ_RAW	-5	
Valuation tax	-1	
conviction_raw	-6	
fs_quality_base	37.5	50 + (-5 × 2.5)
fs_score	~38.5	After OEY 6.4% (income strong ≥3% → +5) and PE pctile adjustments
fs_class	moderate_low	Score 38.5 → moderate_low (between 25 and 40)
FS cap applied?	No cap needed	conviction -6 is already below the +4 cap
FINAL conviction_score	-6	HARD EXIT if holding. CANCEL BUY on any signal.

9.4 Yield Trap Check

Condition	Value	Threshold	Fires?
Z-score (yield vs 5Y own history)	2.00	Must be >1.5	YES
Market absolute threshold (TSX)	7.3% yield	7% (TSX)	YES — marginally above
FCF dividend coverage	0.87x	Must be ≥1.2x	YES
YIELD TRAP RESULT			■■ FIRES — hard gate engaged

■■ BUY signal → CANCEL BUY (yield trap hard gate overrides conviction score entirely)

■■ SELL signal → HARD EXIT (yield trap fires on sell side too)

Note on OEY: OEY 6.4% looks attractive on the surface. But FCF is suppressed by a capex cycle expected to last 3-5 more years. Normalised OEY is closer to 4-5%. More importantly: 13% of the dividend is debt-funded — the real sustainable yield on today's business economics is lower than 7.3%.

10. Operational Guide

10.1 Morning Cron — Pre-Market Sequence

Step	Function	What it does	Time
1	check_earnings_triggers(overrides_map)	Checks earnings calendar JSON. Fires full_recalculation() for passed dates.	2-5 min per trigger
2	run_daily_universe(manifest)	Calls daily_update() for all 200+ assets. ONE fast_info call per ticker. Returns alert_map {ticker:[flags]}.	~20 seconds
3	generate_daily_report(alert_map, manifest)	Formats Fundamental Alerts for Claude daily briefing. Universe health + conviction distribution + flagged tickers.	<1 second

Crontab (6am Mon–Fri):

```
0 6 * * 1-5 /usr/bin/python3 /path/to/conviction_engine_v5_final.py --cron
```

10.2 Signal Timeframe — Definitive Explanation

This is a common source of confusion. The definitive statement:

When this document says 'short-term signal' or 'long-term signal', this refers to a signal fired by a quant model function running on a SHORT interval (e.g. INTERVAL_DAILY — daily candles, produces signals that play out over days to weeks) versus a LONG interval (e.g. INTERVAL_MONTHLY — monthly candles, produces signals that play out over months). The signal_timeframe parameter in modify_signal() is set by the quant model function based on its own interval parameter. The conviction engine does NOT determine the timeframe — the quant model function determines it and passes it in.

Per-function timeframe mapping — from FUNCTION_CONFIG in trade_arrival_analysis.py:

Function (FUNCTION_CONFIG key)	Source file	Interval (from code)	signal_timeframe
OSCILLATOR_DELTA	divergence_comp.py	INTERVAL_DAILY (fixed)	'short'
TRENDPULSE	trendline.py	All intervals — D/W/M/Q/Y	'short' if D/W; 'long' if M/Q/Y
BASELINEDIVERGENCE	general_divergence.py	INTERVAL_WEEKLY (fixed)	'short'
FRACTAL_TRACK	fib_ret.py	D/W/M/Q (selectable)	'short' if D/W; 'long' if M/Q
BAND_MATRIX	bollinger.py	All intervals (selectable)	'short' if D/W; 'long' if M/Q/Y
DELTADRIFT	distance.py	All intervals (selectable)	'short' if D/W; 'long' if M/Q/Y
SIGMASHELL	sigma.py	INTERVAL_DAILY (fixed)	'short'
PULSEGAUGE	sentiment.py	INTERVAL_WEEKLY (fixed)	'short'
ALTITUDE_ALPHA	new_high.py	All intervals (selectable)	'short' if D/W; 'long' if M/Q/Y
COMBINED_STRATEGY	combined_strategy.py	All intervals (selectable)	Set per underlying sub-function

10.3 Integration Point — trade_arrival_analysis.py IS the dispatcher

trade_arrival_analysis.py already acts as the signal orchestrator: it runs all registered functions and collects their signals into signallist[]. This is the correct integration point — one location, not 10 separate files.

Step 1: Create conviction_dispatch.py in the project root:

```
from conviction_engine_v5_final import modify_signal, format_signal_table

# These match the interval constant strings used in the codebase
LONG_INTERVALS = {'Monthly', 'Quarterly', 'Yearly', '1mo', '3mo', '1q', '1y'}

def conviction_dispatch(symbol, signal_value, interval, function_name,
strength=0.75, near_stop=False):
    if signal_value == 0: return None
    direction = 'BUY' if signal_value == 1 else 'SELL'
    timeframe = 'long' if interval in LONG_INTERVALS else 'short'
    try:
        result = modify_signal(ticker=symbol, technical_signal=direction,
```

```

signal_timeframe=timeframe, signal_strength=strength,
quant_model_name=f'{function_name}_{interval}',
long_position_near_stop=near_stop)
print(format_signal_table(symbol, direction,
quant_model=f'{function_name}_{interval}',
signal_timeframe=timeframe, signal_strength=strength))
return result
except RuntimeError as e:
print(f'[conviction] {e} – run full_recalculation("{symbol}") first')
return None

```

Step 2: In trade_arrival_analysis.py — add import + single call:

```

# At top of trade_arrival_analysis.py:
from conviction_dispatch import conviction_dispatch

# In the signal collection loop where 'signallist.append(signal)' exists:
# ADD THIS immediately after signallist.append(signal):
conviction_dispatch(
symbol = symbol,
signal_value = signal, # already 1 or -1 at this point
interval = config.get('interval') or interval, # from FUNCTION_CONFIG
function_name = function_name, # key from FUNCTION_CONFIG e.g. 'TRENDPULSE'
strength = win_rate, # use function's own win_rate as signal strength
)

```

That is the complete integration. Two steps: one new file, one added call. No existing logic is changed. All 10 registered functions route through this single point automatically because trade_arrival_analysis.py already iterates through FUNCTION_CONFIG.

10.4 Post-Earnings Checklist — What Ahil Updates After Each Call

Event	Function to call	Key fields to update
Earnings call reviewed	update_overrides(ticker, {...})	deal_delay_flag, revenue_accelerating, divergence_signal, retention_pct
CEO/CFO changed	update_overrides(ticker, {...})	new_ceo_transition=True, ceo_quality_score=new_value, mgmt_alloc_score=new_value
New earnings date announced	set_earnings_date(ticker, 'YYYY-MM-DD')	Cron auto-fires full_recalculation() when date passes
US ticker PE history thin (<20 pts)	fetch_pe_history_macrotrends(ticker, slug)	Auto-called in full_recalculation if <20 pts and US ticker
Non-US PE history needed	Edit store JSON directly	Set pe_20y_array via Gurufocus / TIKR / Screener.in / manual
New asset in universe	full_recalculation(ticker, overrides)	All fields populated. Run once then daily runner keeps warm.

10.6 CLI Quick Reference

Command	What it does
python conviction_engine_v5_final.py KXS.TO BUY	Snapshot + signal table for KXS BUY
python conviction_engine_v5_final.py --cron	Full morning sequence
python conviction_engine_v5_final.py --report	Daily report only
python conviction_engine_v5_final.py --init universe.txt	Day 0 initialisation
python conviction_engine_v5_final.py --set-earnings KXS.TO 2026-07-15	Set earnings date
python conviction_engine_v5_final.py --overrides KXS.TO	Print current overrides
python telus_test.py	Run TELUS mock test (no network needed)

End of document. Strategy questions → Rohit Malhotra. Implementation → Ahil Khan.